**Estimation: OLS, IV and the first stage**

*Controls: Analysis Method; Approach; Density measure.*

**1. The relationship being estimated.** For a county or MSA $g$, let $Q_g$ be total GDP, $N_g$ total employment and $A_g$ land area. With all industries, the unadjusted outcome is $y_g = \log(Q_g/N_g)$. A sector choice replaces both outcome components by sector values; schooling and college options further adjust $y_g$. Under average employment density, the regression is

$$y_g = \alpha + \rho \log(N_g/A_g) + W_g'\delta + u_g.$$

Here $W_g$ contains the selected controls and fixed effects, $\delta$ their coefficients, and $u_g$ the unexplained part of productivity. The coefficient $\rho$ is an elasticity: an estimate of 0.05 means that 1% greater employment density is associated with about 0.05% higher GDP per employment, holding the controls fixed. The Employment approach replaces log density by $\log N_g$; that coefficient refers to employment rather than density.

**2. OLS.** Ordinary least squares chooses coefficients to make the squared regression errors as small as possible. It uses the observed variation in modern density or employment after accounting for the controls. A positive coefficient need not be a causal density effect: productive places may attract employment, and omitted factors may influence both variables. The app retains availability of the selected historical instrument when forming the OLS sample, making comparisons with IV easier.

**3. IV and the first stage.** Instrumental variables uses the historical instrument $z_g$ to predict the modern employment measure. For average density, the first stage is

$$\log(N_g/A_g) = \pi_0 + \pi_z z_g + W_g'\pi_W + v_g.$$

Two-stage least squares (2SLS) uses this predicted variation to estimate the productivity equation. A causal interpretation requires the instrument to predict the modern employment measure and to affect modern productivity only through the modeled channel, conditional on the controls. Historical timing alone does not establish these assumptions. First-stage Regression displays the predictive relationship, on the sample usable by the selected IV outcome.

**4. When the CH index is selected.** For MSA and state CH models, the index itself depends on the unknown elasticity. OLS therefore runs nonlinear least squares; IV runs nonlinear GMM using the historical instrument. A separate first-stage regression is not offered for CH. The reported CH elasticity is $\rho = \theta - 1$. Selecting a different estimator can change both the point estimate and its uncertainty; compare the reported samples as well as the coefficients.